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ECE 414 Senior Design Project Proposal

Team #: 20

Project Title: **SolarVision Mower: A YOLO and LIDAR-Based Navigation System for Autonomous Solar Powered Lawnmowers**

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Abstract

The Solar Vision Mower is an autonomous, solar-powered mower that makes lawn care sustainable and easy for older adults and individuals with physical disabilities. This network combines a photovoltaic charger dock, object detection based on the YOLO model, and LiDAR-based navigation to achieve efficient mowing without collisions. It is styled after the Raspberry Pi or NVIDIA Jetson Nano, which is a processing unit, with the help of a self-designed battery management system, allowing the unit to be charged independently using solar energy. Real-time monitoring and manual override provision can be achieved using a mobile application. Mechanical design features include Mecanum wheels and adaptive mowing height control. The SolarVision Mower will combine renewable energy, edge computing, and AI-based perception to reduce its environmental impact and increase accessibility. The product will target environmentally conscious homeowners, the ageing population, and the sustainable landscaping market to identify credible, grid-independent robotic mowing products.

Keywords: Autonomous lawnmower, solar power, YOLO object detection, LiDAR navigation, obstacle avoidance, smart agriculture, robotic mowing systems, sustainable energy, edge computing, and human-centered design.

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Introduction

Background

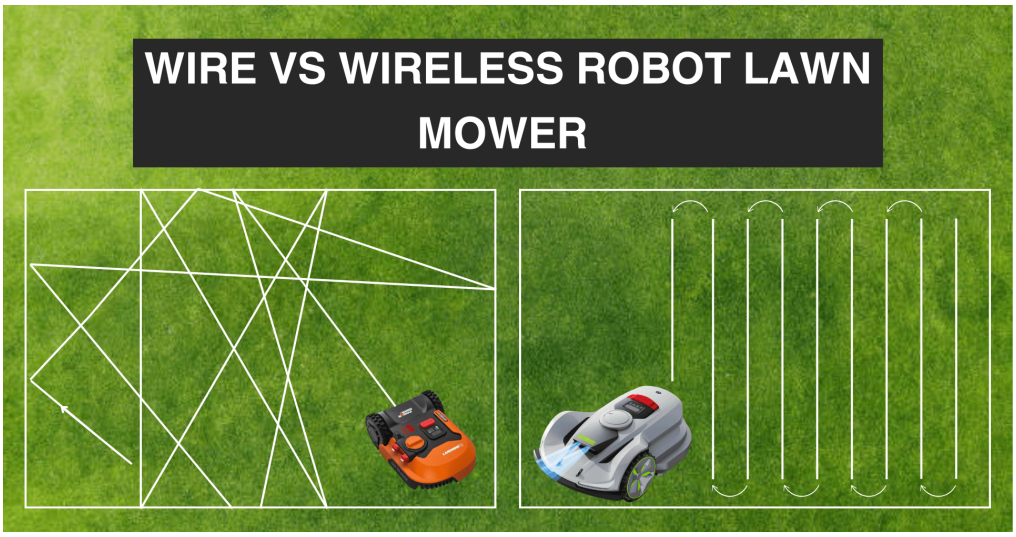
Autonomous outdoor service robots have emerged as one of the most promising applications of robotics, which combine artificial intelligence (AI) and renewable energy integration. Among these, robotic lawnmowers have gained attention in the past 3 decades. These tools provide convenience, save time, and enhance safety as opposed to manual mowing. The earliest robotic mowers, Husqvarna Automower, were introduced in the 1990s and significantly improved throughout the next 2 decades [1]. These early auto-mowers relied on perimeter wires installed around a lawn to define boundaries and guide navigation, requiring an intensive setup [2]. They lacked sophisticated obstacle detection techniques and operated on simple random mowing patterns. These attributes limited efficiency and adaptability.

In recent years, there have been advancements in computer vision, sensor fusion, and renewable energy solutions. These technologies have paved the way for more intelligent, sustainable lawn care systems. For instance, there are rapid improvements of deep learning-based models such as YOLO (You Only Look Once) for real-time object detection [3]. Using YOLO, robots can now identify and classify obstacles, including people, pets, or garden objects, far more reliably than traditional rule-based approaches [3]. At the same time, Light Detection and Ranging (LiDAR) sensors are widely used in autonomous vehicles. This technology enables precise mapping and navigation by creating detailed three-dimensional spatial representations [4].

Apart from sensor innovations, renewable energy has also become central to consumer robotics. The integration of solar energy into autonomous devices is a significant energy and sustainability solution. It offers a pathway toward reducing dependence on manual charging and

fossil fuel-powered garden tools [5]. A solar-powered system ensures sustainability, reduces operational costs, and lowers carbon emissions. This ensures alignment with the global efforts to decarbonize small-scale machinery [6,7,8].

Figure 1: Evolution of robotic lawnmowers from wired navigation to sensor-based intelligent systems [9].



Problem Statement

Despite progress, existing robotic lawnmowers continue to face several limitations. As noted earlier, many commercially available models depend on wired perimeter systems for navigation. These systems increase installation cost and complexity [10]. Others rely solely on GPS, which may suffer from inaccuracies in residential environments with tall buildings or dense trees [11]. Furthermore, consumer robotic mowers typically operate on rechargeable lithium-ion batteries that require docking at powered charging stations [12]. While efficient, these stations rely on grid electricity, making the systems less sustainable and unsuitable for regions with unstable energy supply.

Another issue is safety. Although robotic lawnmowers often incorporate bump sensors and basic obstacle detection, they remain prone to accidents according to a European study that evaluated their impact on Hedgehogs [13]. This finding implies that the mowers can harm

children, pets, or fragile garden furniture. This raises ethical concerns about user safety, liability, and the responsible deployment of autonomous garden robots.

From a market perspective, the affordability of advanced robotic lawnmowers remains a barrier. Premium models with advanced navigation features often exceed USD 1,300 [14], while Selvanesan *et al*. indicate that the price of production can be as low as USD 130 [10]. This way of pricing hinders the accessibility of smart lawnmowers for middle- and low-income households. For elderly users, who stand to benefit most from autonomous mowing solutions, these cost barriers and technical setup requirements often remain prohibitive.

Our project seeks to address these issues by designing a self-sustaining, solar-powered autonomous lawnmower. This will combine YOLO-based object detection with LiDAR navigation for enhanced safety, precision, and sustainability. Solar charging will ensure that the served population does not have to rely on frequent charging of the lithium-based batteries. A model of the proposed mower is shown below.

Figure 2: Proposed SolarVision Mower and its docking deck models

Objectives of the Proposed Project

Primary Objective. The main objective of the SolarVision Mower is to create a functional prototype of an autonomous lawnmower that prioritizes sustainability, accessibility, and intelligent navigation.

Other Objectives. Specifically, the project will aim to:

1. Develop a solar-powered charging system using a compact shed structure that allows the mower to recharge independently, reducing reliance on grid electricity.
2. Implement YOLO-based object detection for real-time obstacle detection, ensuring safe operation around people, pets, and objects.
3. Integrate LiDAR for accurate environmental mapping and navigation, and efficient and adaptive mowing patterns.
4. Create a mobile app for remote control and monitoring, enhancing usability for elderly users and caregivers.
5. Evaluate the economic feasibility and scalability of the system for broader consumer adoption.

By combining renewable energy with cutting-edge AI-driven navigation, this project seeks to move beyond the limitations of existing robotic mowers. At the same time, it focuses on user accessibility, particularly for elderly individuals.

Adopted Approach

The adopted approach integrates multiple technological domains into a cohesive system. The mower's driving function will run on a compact onboard computer, using an STM32N6 microcontroller from STMicroelectronics. The MCU comes with a built-in proprietary Neural Accelerator, the first of its kind from STM, which can operate at 600 GOPS, more than enough

for our use case [15]. LiDAR sensors will generate real-time point clouds of the mowing environment. When these are fused with visual detection, they will allow the system to triangulate obstacles, calculate optimal paths, and maintain consistent mowing coverage. This approach, called Simultaneous Localization And Mapping (SLAM), is not new; yet its pairing with deep learning, and its deployment on an autonomous lawnmower will be fairly novel.

The solar charging system will be implemented through a mini-shed housing 3 monocrystalline photovoltaic panels. The mower will autonomously navigate to the shed when battery levels are low or after completing a mowing cycle. This system not only minimizes manual intervention but also ensures the mower remains functional in off-grid conditions.

For elderly accessibility, a mobile/web app will provide real-time monitoring, scheduling, and manual override options. This app-based control will address usability concerns while giving family members the ability to assist remotely. This measure will enhance accessibility for all populations. This hybrid approach, combining AI-driven perception, LiDAR-based navigation, solar sustainability, and human-centered design, distinguishes the SolarVision Mower from existing robotic lawnmowers that typically adopt only one or two of these technologies.

Current Trends in the Field

The global market for robotic lawnmowers has expanded rapidly. It had a market size of USD 1.5 billion in 2023 [16], and USD 1.15 billion in 2024 [17]. This was estimated to grow up to USD 2 billion by 2030 [16], and USD 3.01 billion by 2032 [17]. The market has an estimated Compound Annual Growth Rate (CAGR) of between 10.9% to 14.8% between 2023 and 2030 [10,16,17]. Much of this growth is driven by consumer demand for convenience and eco-friendly outdoor equipment. Leading manufacturers such as Husqvarna, Bosch, and Worx are incorporating GPS-based navigation and smartphone connectivity into their products [17].

However, the majority of these systems still rely on docking stations powered by conventional electricity.

Figure 3: Global robotic lawn mower market trends [17].

From a research perspective, there is growing emphasis on integrating autonomous navigation technologies from self-driving vehicles into smaller consumer robotics. For instance, researchers have explored LiDAR-GPS fusion for precision agriculture robots [11,12,18,19] and deep learning for real-time obstacle avoidance in mobile robots [20]. The adoption of renewable energy solutions, including solar integration, remains an emerging area, with limited deployment in commercial robotic mowers despite demonstrated potential in academic prototypes [21].

Ethically, robotic mowers raise debates over safety, energy sustainability, and labor displacement [13]. While autonomous mowing reduces physical strain for elderly users, and more generally is convenient for everyone, improper deployment could lead to safety hazards. Manufacturers are therefore under increasing pressure to meet ethical design standards, emphasizing safety certifications, AI transparency, and energy efficiency.

Economically, as battery and solar technologies become cheaper, the cost of sustainable robotic systems is expected to decline [22]. This trend could improve accessibility and

democratize advanced lawn care technologies for broader populations. This will particularly affect regions where manual mowing remains labor-intensive.

Related Real-World Commercial Systems

A review of current commercial systems reveals several benchmarks and limitations. Husqvarna Automower remains a leading product, but it requires perimeter wire installation and relies on traditional charging docks [2]. Bosch's Indego line integrates GPS mapping but has limited obstacle detection capabilities. The Worx Landroid offers smartphone connectivity and scheduling features but, like others, is not solar-powered [23]. However, these existing prototypes offer significant frameworks and highlight the niche on which the proposed mower will be building.

Research prototypes, such as solar-powered agricultural robots for crop management, have demonstrated the feasibility of renewable-powered autonomy [11,12,18,19]. However, the direct application of solar energy in consumer lawnmowers is still underexplored. This gap provides an opportunity for the SolarVision Mower to distinguish itself as one of the first to integrate solar charging, deep learning based vision, and SLAM-based navigation into a unified consumer-ready system.

From a sustainability perspective, our design directly addresses the ethical and economic concerns surrounding energy use and affordability. The proposed mower leverages renewable energy and reduces reliance on costly wired navigation. Therefore, it has potential for both ecological and social impact. While current commercial systems have advanced beyond basic wire-guided models, they remain constrained by energy dependence and limited perception. Additionally, by adopting YOLO and LiDAR for object detection and navigation, our prototype offers a forward-looking solution with the potential to reshape the consumer robotics landscape.

Members Responsibilities and Technical Goals

Each technical goal will be spread out through different time frames for completion. Throughout those time frames, the group as a whole will put in a combined effort, where each component under the major technical goal will be split up among the 4 of us. The major technical goals will be tackled sequentially, starting with the Solar Shed, followed by the Battery Management System for the on-board battery, the autonomous navigation system, and finally the Android app to monitor and control the lawnmower through the phone if need be. Realizing these goals in a timely manner is absolutely essential to the completion of this project, and as such, a more detailed overview is given below.

Firstly, we will build the Solar Shed. The shed itself, which will be built out of wood, will be made waterproof, and covered in 3 series 12V 100W peak output Monocrystalline solar panels. These solar panels will then charge a stationary deep cycle Lead Acid battery pack through a custom Battery Management System (BMS), which will then charge the mower's Lithium Ion 36V 10Ah battery through a second BMS. The Solar Shed will operate as a complete unit in-and-of itself, not attached to the grid's power system. While this does have its drawbacks, such as being unable to charge on cloudy days, we believe that it will bring around a simplicity of installation that won't be garnered by connecting the system to the grid. Additionally, since mowers are typically used infrequently (once every two or three weeks), the 300W output should be more than enough to maintain the mower's state-of-charge.

Next, we will start working on the autonomous mower itself. Or rather, a bench-tested version of it. At this point in time, all the 4 of us will tackle separate tasks within the umbrella of this bench-test; namely, the LiDAR system, the GPS/RTK system, the YOLO vision based object

detection system, and the Sound-based Triangulation system. Each of us will prepare our sensors and their corresponding interfaces, such that when the time to put the pieces together comes around, the sensor system will be sufficiently abstracted away, to the point that we only need to worry about the actual navigation algorithm and final physical design of the mower.

Next, we will split up into groups of 2. One group will be working on the navigation system for the mower, and the other will work on the physical design, and as such, will hold each other accountable to meet our deadlines. As stated earlier, our main design constraints for both the navigation system and the physical design of the mower, is cheapness and reliability. For instance, a turning mechanism for the wheels may prove challenging; rather a 4-wheel drive design with Mecanum wheels may be more cost-effective and reliable, and also provide the benefits of a shorter turning radius [28]. We also plan to build a custom Lithium Ion battery out of 18650 cells in a 10S4P configuration, as this approach drastically reduces the cost compared to buying a ready-made battery pack. Once the body of the mower and the navigation system is complete, we will attempt to create an Android app to control/override the mower remotely, if we have the time.

Below is a bullet point list of how we anticipate our project will pan out. We will be starting it soon this semester, and aim to complete it by the end of next semester.

1. Autonomous Driving (Ideas - not necessarily using them all)
    a. LiDAR
        i. Interface with Arduino
    b. GPS/RTK
        i. Interface with Arduino
    c. YOLO
        i. Interface Arduino with Computer running YOLO
        ii. Run Navigation System on Computer
    d. Triangulation System - TBD
2. Software App
    a. Connection with Arduino - Bluetooth/WiFi

    b. Controls/Overrides Mower if needed
3. Overall Design
    a. *Cost Efficient*
        i. Main idea - *Make it Cheaper and Better than Market!*
    b. Lawnmower Design
        i. Wheel Design Where It Can Drive Vertically and Horizontally
            1. Mecanum wheels.
        ii. Water and Heat Resistance
        iii. Servo Motor Actuated Mower Height
            1. Arduino Based; App Connected
    c. Shed Design
        i. 100W Monocrystalline Solar Panel
        ii. Batteries
            1. Lead Acid fixed battery pack, Lithium Ion on-board battery pack
            2. BMS
                a. Boost Converter and Regulator for the Fixed Lead Acid battery pack (12V → 48V)
                b. Charging System to Charge the Mower
                c. BMS for the On-Board Battery itself
        iii. Lightweight

Projected Community Impact and Relevant Ethical Issues

The proposed project will have significant benefits for the community. First, it will be a resource for the elderly and people living with disabilities. These community members often struggle to do yardwork, and sometimes they can not do it because it requires considerable physical effort [24]. Because of its automation features, the lawnmower will improve the quality of life, promote independence, and reduce the cost of lawn care.

Besides, the lawnmower will protect the environment from fossil fuel damage as it relies on solar power. Lawnmowers that are powered by gas pollute the environment through carbon dioxide and particulate matter emissions [25]. Solar energy and efficient batteries for power storage ensure that this project lowers the carbon footprint and promotes sustainability [6,7,8].

Automated mowing also ensures that lawns stay healthy at all times, supporting the sequestration of carbon and biodiversity [11].

Additionally, the automated lawnmower makes resource management easier. The system utilizes LiDAR-based navigation and YOLO object detection techniques. These ensure it avoids unnecessary mowing, prevents collisions, and reduces energy waste. [3,4,15]. The proposed mower, therefore, aligns itself with the current and future trends in farming and landscaping, which are both innovative and sustainable.

Despite the benefits highlighted above, the YOLO lawnmower project raises some ethical issues that must be addressed. The first issue is safety around residential areas. Automated machines can harm children, pets, or local wildlife [13,26]. Therefore, they must be adequately engineered with safety measures. For the proposed project, thorough testing and the use of fail-safe systems will help improve safety. In addition to safety, automated mowers also pose the risk of displacing human labor [27]. The likelihood of causing job loss could lead to project resistance. Therefore, educating community members on the benefits of the automated mowers will be necessary.

This project seeks to balance technological advancement and social responsibility. It prioritizes safety, accessibility, and sustainability at its core. This way, community members will reap maximum benefits while the project avoids ethical risks.

Relevant Engineering Standards, Project Constraints, Risk, and Uncertainty

1.1 Engineering Standards

A. IEEE 1547-2018: Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces

a. IEEE 1547 establishes criteria for integrating distributed renewable energy systems, such as photovoltaic (PV) units, into existing grids. Although the SolarVision Mower functions as an off-grid system, this standard is relevant for its solar charging module. This ensures proper design of the charge controller, inverter, and safety cutoff mechanisms [29]. Following these guidelines promotes energy efficiency and safe operation under variable sunlight conditions.

B. **IEEE 1028-2008: Standard for Software Reviews and Audits**

a. Because the mower's autonomous functions rely heavily on YOLO-based object detection and LIDAR sensor fusion, the software quality directly affects safety and reliability. IEEE 1028 provides procedures for systematic software verification, design reviews, and audits [30]. Implementing this standard ensures that code modules related to image recognition, sensor fusion, and control logic undergo formal inspection and testing. This reduces the likelihood of undetected software bugs or faulty decision-making algorithms.

C. **IEEE 61508-2010 / IEC 61508: Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems**

a. This is a foundational international standard that governs functional safety in autonomous and robotic systems. It defines requirements for system reliability, hardware redundancy, and response to hazardous situations [31]. For the SolarVision Mower, compliance means integrating fail-safe mechanisms. These include emergency stop sensors, automatic power cut-off when detecting human presence, and obstacle avoidance algorithms validated under safety integrity

levels (SILs). Using functional safety guidelines ensures the robot minimizes the probability of injury or property damage in unpredictable outdoor environments.

D. **IEEE 802.11: Standard for Wireless LAN Communication**

a. Because the mower can be remotely controlled and monitored through a mobile application, reliable wireless communication is critical. IEEE 802.11 provides protocols for Wi-Fi communication, ensuring secure and interference-resistant data transmission [32]. Adhering to this standard supports smooth user-device interaction. It will boost remote start/stop commands, diagnostic data upload, and firmware updates.

E. **ISO 8373:2021: Robotics – Vocabulary**

a. This ISO standard defines terminology and basic concepts used in robotics [33]. Although the standard was withdrawn, it is essential for establishing a shared understanding within the design team and aligning with research literature. By classifying the mower as a "mobile service robot," the team can apply appropriate risk assessment and operational design standards as recommended by ISO.

1.2 Major Categories of Project Constraints

(a) Technical Constraints

The SolarVision Mower integrates multiple advanced technologies, each with its own limitations. The accuracy of object detection is restricted by lighting conditions and the processing power of the embedded computing system. For example, the YOLOv5 model performs best under stable illumination; shadows or reflective surfaces can lead to false detections [15]. Additionally, LIDAR sensors may exhibit reduced precision in heavy rain or

when the laser signal is scattered by tall grass [4]. Thus, our project will include simple weather detection, to find the best times to mow a lawn.

The battery capacity also imposes a significant technical limit. To sustain mowing operations and sensor processing, the power management system must efficiently battery storage, and energy consumption. Furthermore, autonomous navigation requires a real-time operating system (RTOS) capable of handling multiple data streams simultaneously without lag [34]. Any hardware-software mismatch could cause delays that affect path accuracy and obstacle avoidance. Thus, for safety reasons, our mower will utilize *normally open* relays/switches between the power supply and all major loads (motors). This will ensure that upon loss of power, or a software triggered signal, the motors can be forced to shut off.

Lastly, safety redundancy is technically demanding. The mower must incorporate multiple sensors (infrared, ultrasonic, and mechanical bumpers) to ensure that even if one fails, others still provide fail-safe feedback. Implementing these redundancies increases complexity, testing time, and cost. However, it is necessary for compliance with IEC 61508 safety standards [31].

(b) Economic (Budget) Constraints

Budget constraints are particularly relevant for a prototype project built using student or institutional funding. High-quality LIDAR sensors (such as Velodyne or Ouster) and NVIDIA Jetson-class processors can significantly increase costs. A single industrial-grade sensor can exceed USD 1,000. Reliable lithium-ion battery packs and photovoltaic modules further increase expenses [28].

To remain cost-effective, the design uses open-source machine learning frameworks and affordable hardware alternatives such as RPLIDAR for distance sensing and the novel STM32N6 (currently $20 on Mouser Electronics) for vision inference. Even then, balancing performance versus cost remains a challenge. Low-cost components may have limited durability or lower precision.

Economic sustainability is also tied to manufacturing scalability. While the prototype may operate reliably, commercial production would demand cost optimization in materials, mass manufacturing, and supply chain logistics [35]. Thus, cost-benefit analysis must continue beyond the prototype phase to determine whether the mower can compete with existing models.

(c) Environmental and Sustainability Constraints

The environmental motivation behind this project is strong. It aims at reducing emissions from gas-powered lawnmowers and supporting sustainable home automation. According to the U.S. Environmental Protection Agency (EPA), small gas engines produce a disproportionate share of lawn care emissions. This contributes to carbon monoxide and volatile organic compounds [36]. The SolarVision Mower eliminates fossil fuel dependence through solar recharging and low-voltage DC operation, directly reducing the carbon footprint.

However, environmental constraints also involve energy storage, material choice, and disposal. Lithium-ion batteries, while efficient, pose recycling challenges and potential fire hazards if mishandled. To comply with sustainability guidelines, the design emphasizes modularity, allowing batteries to be replaced or recycled independently. Solar panels must also meet durability and efficiency standards under outdoor exposure.

Weather variability presents an inherent environmental limitation. The mower's solar charging rate depends on sunlight availability, meaning overcast conditions may require the system to postpone mowing or limit its operational window. The autonomous scheduling algorithm must therefore integrate weather forecasts to optimize mowing intervals.

1.3 Risk and Uncertainty Assessment

Risk Table

Technical Goals	**Risk Level**		
	Low	**Moderate**	**High**
Object Detection (YOLO)		The model misclassifies obstacles in poor lighting or tall grass. Mitigation: Train model with diverse, perhaps custom datasets; integrate LIDAR depth confirmation.	
Sensor Fusion			LIDAR and camera data desynchronization. Mitigation: Implement timestamp synchronization and real-time data buffering.

Battery Management			Overcharging, overheating, or premature degradation. Mitigation: Add temperature sensors and overcurrent protection circuitry.
Solar Charging Dock		Inefficient charging during cloudy weather. Mitigation: Include backup plug-in charging option and energy storage estimation algorithm.	
Software Reliability			Unexpected crash or algorithm loop due to memory overload. Mitigation: Follow IEEE 1028 reviews and continuous integration testing.
Mechanical Design			Blade collision or motor failure under uneven terrain. Mitigation: Add automatic shutdown sensors and reinforced housing.
Network Communication		Wi-Fi connectivity failure causing loss of remote control. Mitigation: Integrate fallback manual override and local control mode.	

Environmental Impact	Battery disposal or PV panel waste. Mitigation: Use recyclable materials and certified disposal vendors.		
Legal / Safety Compliance			Injury or property damage during autonomous operation. Mitigation: Comply with IEC 61508 functional safety protocols and emergency stop design.
Economic Feasibility	Prototype costs exceed projected budget. Mitigation: Utilize low-cost open-source components and staged development milestones.		

Discussion of Risks and Mitigation

I. Technical Risks

A. The most critical risks arise from sensor reliability and navigation accuracy. Because the mower relies on visual data, conditions like dusk, rain, or dense foliage can impair YOLO's accuracy. To mitigate this, multi-sensor fusion will be employed, combining visual recognition with depth sensing to ensure redundancy [37]. The risk remains moderate, as environmental testing may reveal new edge cases.

II. Economic Risks

A. The project's funding limits influence hardware selection. A low-budget prototype may underperform compared to commercial models. This will potentially lead to underestimation of cost-to-performance ratios. To control this, the design follows an incremental approach. First, we will develop a proof of concept before scaling. Additionally, using modular parts allows easy replacement without rebuilding the entire system.

III. Safety and Legal Risks:

A. Autonomous mowing poses serious safety concerns, especially in public or residential areas. The risk level is high if fail-safe mechanisms are not rigorously tested. To meet functional safety standards (IEC 61508), the mower will feature multiple safety layers [31]. These include proximity sensors, AI-based human detection, and an emergency stop button accessible both physically and via the mobile app.

IV. Environmental Risks

A. While solar power reduces emissions, weather dependency introduces operational uncertainty. Extended cloudy days may delay mowing schedules, creating user dissatisfaction. This is mitigated through adaptive scheduling, where the system monitors battery charge and predicts available solar energy. Additionally, recyclable components and minimal use of plastics reduce environmental footprint.

V. Manufacturing and Sustainability Risks

A. Transitioning from prototype to large-scale production introduces uncertainty in cost and durability. Early design documentation and compliance with IEEE/IEC standards will ease this transition by providing traceable validation steps. Sustainable design choices align the mower with circular economy principles.

1.4 Uncertainty Factors and Future Considerations

Uncertainty in this project stems from technological evolution and external environmental factors. Advances in AI vision algorithms, for instance, may soon make the YOLO model obsolete or less efficient compared to emerging transformer-based architectures. Thus, the system is designed with software modularity to allow for future upgrades without hardware replacement.

Another uncertainty lies in solar energy availability, which is variable across seasons and geographical regions. Long-term performance assessment under different latitudes and climates will be necessary to validate system reliability. Similarly, the long-term degradation of photovoltaic panels and batteries introduces operational unpredictability that can only be measured through field testing over several months.

Legal and ethical uncertainties also exist. The mower must comply with local regulations concerning autonomous machinery in residential areas. For instance, some municipalities require geofencing to prevent robotic devices from crossing property boundaries. Public perception and acceptance may further influence adoption rates, particularly concerning safety and job

displacement fears in the landscaping sector [28]. Despite these uncertainties, the SolarVision Mower remains a promising contribution to sustainable automation.

Design methodology, prototyping, validation, and hardware block diagram

For the purposes of this project, we created a diagram (and approximate timeline) of how we expect this project to flow. As mentioned in the Technical Goals section, we aim to tackle large portions of the project as a group, and then subdivide said portions between us. As such, we will begin (starting in November) to build a Solar Shed, which will house the completed mower.

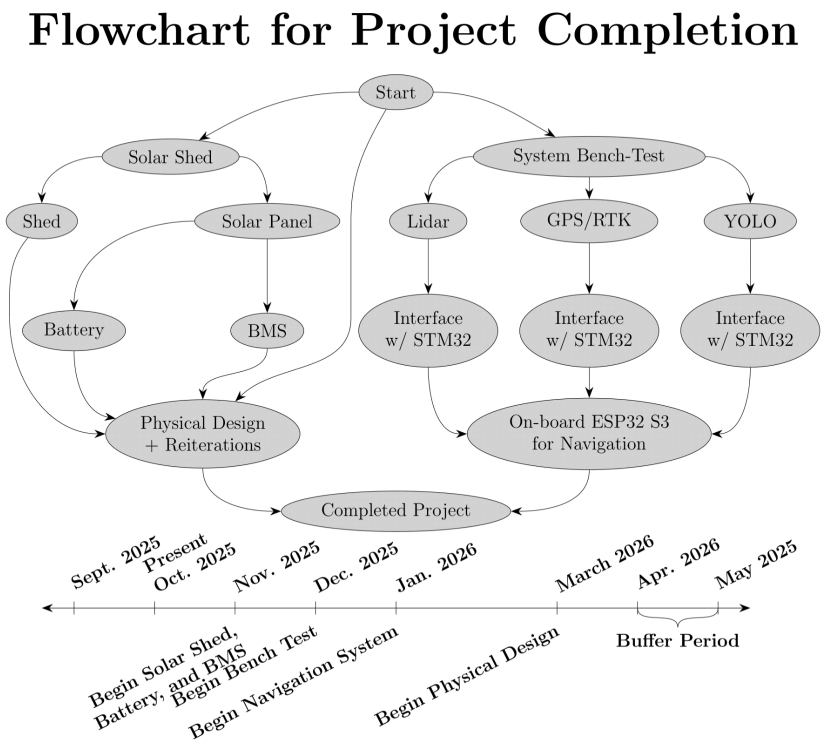
Our hope is that the bench testing, prototyping, and design iterations of the mower itself will take much more time than designing the Solar Shed. As such, we've decided to begin *bench testing* the components of the project by <u>this December</u> at the latest. This bench test entails the following:

- Individually testing each sensor/motor component that will be used (LiDAR, GPS/RTK, YOLO, drive motors, and the height adjustment motor).
- Creating a *modular software interface* that will ultimately allow us to abstract the hardware away by the time we get to the next phase: Autonomous Navigation.

Once the bench testing and interfacing is complete, we will transition to the *Autonomous Navigation* phase of our project. In this phase, we will begin the software development for the navigation of the mower. This entails the following:

- Utilizing the previously established interfaces to effectively communicate with the hardware components.

- Writing software to locate the mower in 3d space, map out the given lawn, and run an optimal path algorithm to mow the entire lawn with the least distance travelled.



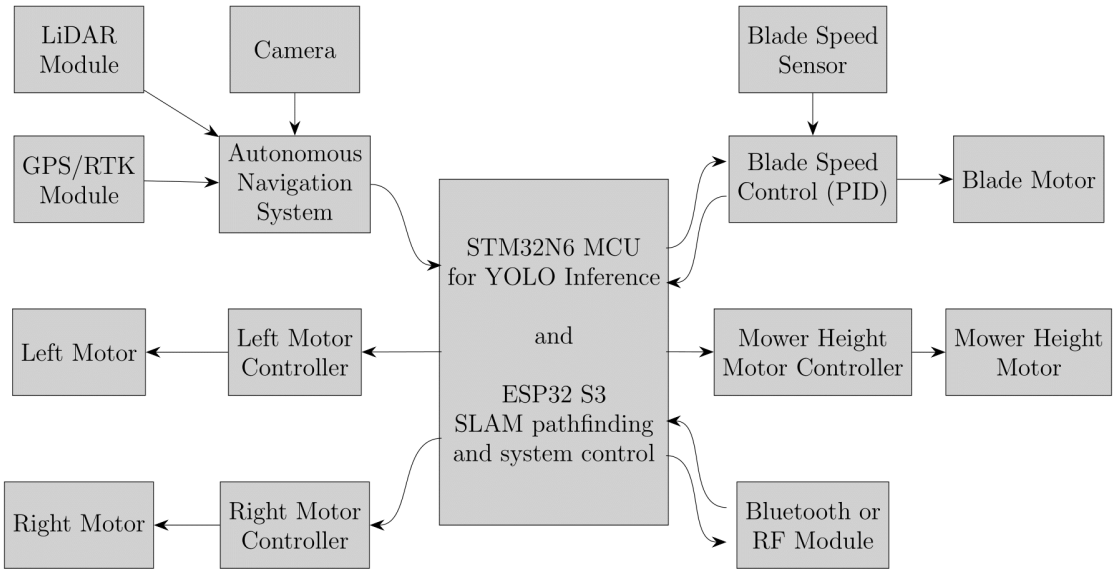
For the hardware side of our project, we've decided to use the following components:

- Sensors:
    - Lidar: Adafruit-Slamtec RPLIDAR A1
    - Camera: Standard USB Logitech Camera
    - GPS: Adafruit 66 channel GPS Module
    - Blade speed sensor: Custom built with Infrared Sensor
    - On-board STM32N6 SBC
- Motors:

- Blade Motor: 200W DC Motor
- Mower Height Motor: Standard 12V Linear Actuator
- Left and Right Motors: Geared 12V 250W Motors

- Battery Module
  - Custom built 36V 10Ah 18650 battery pack
  - 10S4P Battery Management System
- Inference Engine and System Controller:
  - Inference: STM32N6 SBC, with C++ YOLO inference engine, utilizing NPU
  - System Controller: ESP32 S3, for SLAM object detection and pathfinding.

Block Diagram of Autonomous Lawnmower



Relevant Technical Background

Selected Technical Articles

***Article 1*: The YOLO Framework: A Comprehensive Review of Evolution, Applications, and Benchmarks in Object Detection**

Authors: M. L. Ali and Z. Zhang

Journal: *Computers*, vol. 13, no. 12, p. 336, 2024.

DOI: 10.3390/computers13120336

Link: https://doi.org/10.3390/computers13120336

ISBN: ISSN 2073-431X

Here, the review paper will compare the evolution of the YOLO (You Only Look Once) model between YOLOv1 and YOLOv8, as well as the architectural upgrades that enable real-time object detection on embedded and edge devices. The article highlights YOLOv5 and YOLOv7 as applicable in autonomous mobile systems that require low-latency inference, such as robotics and intelligent surveillance. It also demonstrates the performance on the benchmarks of COCO and Pascal datasets, which confirms the accuracy-speed tradeoff of YOLO when compared to R-CNN and SSD approaches [38].

Relevance and Comparison

The SolarVision Mower advertisement utilizes YOLOv5 as its primary object detection model for categorizing real-time obstacles in real-time. Unlike the general-purpose applications, the architecture of YOLO in this project is tailored to the requirements of an outdoor environment, where LiDAR point clouds and the computational constraints of solar power are present [38]. It is new, as it utilizes visual detection with LiDAR-based spatial awareness to autonomously mow the lawn in a design that has not previously been optimized for outdoor renewable-powered robotics.

***Article 2*: LiDAR Technology for UAV Detection: From Fundamentals and Operational Principles to Advanced Detection and Classification Techniques**

Authors: U. Seidaliyeva, L. Ilipbayeva, D. Utebayeva et al.

Journal: *Sensors*, vol. 25, no. 9, p. 2757, 2025.

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The principles of LiDAR systems, which the authors will discuss in this paper, include distance measurement based on the time-of-flight principle, angular resolution, and object classification. It illustrates that LiDAR is applicable in 3D mapping and tracking motion, especially with UAVs that require adequate environmental modeling [39]. The scientific literature review discusses scanning, point cloud filters, sensor fusion, and computer vision models. This is designed to enhance scanning in various lighting conditions and weather.

Relevance and Comparison

The SolarVision Mower follows the same principle as LiDAR, aiming to work efficiently in diverse ground conditions through the use of technology. The LiDAR to be utilized in this project is modified to scan the surrounding environment and dynamically avoid objects in a residential area. The LiDAR and YOLO-based system also consumes less energy, and it is less sensitive to weather, which is attributed to its solar power usage and low-power processors. This is because it has been able to operate with reduced discretion and, consequently, can work on a smaller scale [39].

***Article 3*: Transforming Edge AI: The power of neural processing units in modern microcontrollers**

Authors: White paper from STMicroelectronics

Link:https://content.st.com/whitepaper-transforming-edge-ai-with-npus-in-microcontrollers-z11-5780.html

AI/ML has come a long way in the last few decades, and as of late, deep learning on the edge has grown substantially. Traditional general purpose processors are capable of running these advanced models, but not at any scale needed to meet demand. The authors of this whitepaper thus introduced STM's proprietary Neural-ART Accelerator, capable of increasing inference speed up to 127x when compared to CPU-only ARM Cortex-M55s (when running INT8 quantized and pruned Tiny YOLOv2). This enables, for the first time, inexpensive object detection at up to 32 fps, running on a fraction of a watt.

Relevance and Comparison

This directly relates to the SolarVision Mower, as it will contain an STM32N6 at its core, running a YOLOv8n model directly on the MCU's Neural Accelerator. This design dramatically improves the safety and reliability of the mower, as a higher throughput at the image processor directly correlates with a higher accuracy during path planning and object detection.

8.2 Selected Patents

***Patent 1*: Autonomous Lawn Mower and Method for Operating the Same**

Patent No.: US 11,215,987 B2

Inventors: J. Wahl, H. Björkman, and L. Eriksson

Assignee: Husqvarna AB

Publication Date: Jan. 4, 2022

Abstract and Description

This patent describes a robotic lawnmower equipped with perimeter wire navigation and a collision sensor. The mower will be guided along a fixed, predetermined route and will charge its batteries at a grid-based charging point. Though location-bound and without computer vision, it is efficient [40].

Comparison

The SolarVision Mower utilizes LiDAR-based mapping and YOLO-based visual perception, eliminating the need for wired boundary navigation.

***Patent 2*: Solar-Powered Automated Grass Cutting Robot**

Patent No.: US 10,941,872 B2

Inventors: K. Takahashi and Y. Fukuda

Assignee: Honda Motor Co., Ltd.

Publication Date: Mar. 9, 2021

Abstract and Description

A photovoltaic-powered robot mower charges a lithium battery during operation. Its navigation utilizes infrared and proximity sensors to avoid obstacles [41]. Although the patent can unite renewable energy and automation, it lacks a machine-learning-based perception that limits the distinction of obstacles and adaptive path planning.

Comparison

The SolarVision Mower is even broader as it merges solar energy with the integration of YOLOv5 and LiDAR to conduct intelligent classification and environmentally responsible navigation. The ability of a mobile application to monitor and modify the schedule also enhances the ease and complexity of the automation for customers.

***Patent 3*: LiDAR-Based Collision Avoidance for Autonomous Ground Vehicles**

Patent No.: US 11,764,001 B1

Inventors: B. Yang, H. Kim, and S. Huang

Assignee: Boston Dynamics Inc.

Publication Date: Sep. 12, 2023

Abstract and Description

This patent introduces a LiDAR-based collision-avoidance system that applies to real-time mapping on a 3D map and predictive motion estimation to prevent collisions with obstacles. The technology is primarily intended for use in industrial/defense robots [42]. It is involved with autonomous navigation under the dynamically changing conditions, multi-sensor integration, and spatial segmentation.

Comparison

In the SolarVision Mower advertisement, the same LiDAR-associated safety concepts are applied, but scaled down to the realm of home automation. Precision and energy efficiency are also balanced in the mower, compared to small-scale, solar-limited operations, which focus on high-speed land traversal [42]. The originality of the project lies in the integration of LiDAR-YOLO fusion within the scope of renewable energy, and the design has also been adjusted to support the needs of affordability and safety in the consumer environment.

8.3 Technical Summary and Synthesis

The technological basis of the SolarVision Mower design comprises all the literature reviewed and patents. The scientific community confirms that deep learning (YOLO) and LiDAR systems are already advanced enough to incorporate robots at a low price [39]. Until that day, the existing patents [40,41,42] indicate that commercial mowers remain dependent on grid

charging, wired boundaries, and simplistic sensors. SolarVision Mower has a distinctive feature of a combination of three innovations:

i. Solar independence: complete charge of solar, independent of the grid.
ii. Hybrid perception: The LiDAR depth image and YOLO real-time object detection are used to classify the obstacles.
iii. Edge intelligence: real-time decision making using low-power computing based on the Raspberry Pi or Jetson Nano.

Together, the innovations address the key weaknesses of the current commercial and patented products: the lack of sustainability, the inadequate perception, and barriers to entry. The SolarVision Mower is therefore not just a plus in the technical and ethical aspects (it is safer, has a smaller carbon footprint, and makes older people more independent with intelligent, renewable, and affordable automation), but also an advantage to the beneficiaries of the product themselves.

Marketability and Lifecycle

SolarVision Mower has high market potential as there is an increased trend of consumers moving towards outdoor technologies that are sustainable and autonomous. It is estimated that by 2030, the global robotic lawn mower market would reach USD 4 billion and its compound annual growth rate (CAGR) would be around 10-14 percent as more owners of houses purchase innovative and environmentally friendly solutions [32]. The combination of renewable energy and IoT technologies supports this tendency in the market, as systems of automated processes supported by solar energy have significantly increased their popularity in response to the increased energy cost and growing awareness of environmental issues.

The SolarVision Mower is the only mower with a solar source of power as its primary energy source, and it does not depend much on the electrical system, which reduces the

operational costs. The rival products (Husqvarna Automower and Bosch Indego) are charged on a battery with household electricity. In comparison, the photovoltaic-based charger and a modular structure of SolarVision enhance cost-efficiency, lower the lifecycle emissions, and attract sustainability-conscious communities and homeowners [30].

Economically, a cost of USD 400-600 per unit (estimated production cost of SolarVision) is reached by utilizing inexpensive and open-source components (e.g., Raspberry Pi, LiDAR, and solar PV modules), which allow the company to sell at USD 1,000-1,200. This is being competitive in the high-end home automation market, besides encouraging environmental accountability. The intended audience consists of the middle- and high-income families and communities, retirees, and intelligent housing developers with a focus on sustainable planning [29].

According to a lifecycle assessment (LCA) framework, the SolarVision Mower is based on the principles of the circular economy, in which the components are modular, recycled, and easy to maintain. Solar panels, motors, and battery systems can be easily replaced by themselves, which contributes to the increased life of the products, as well as reduces waste [31]. As per the Pyramid of Production Systems model, the mower passes through the design phase, production phase, usage phase, and recycling phase with a low environmental footprint. The estimated product life cycle is 7-10 years, where a software update is to project the product lifespan further [33].

Altogether, SolarVision is a profitable, sustainable, and socially responsible innovation. It combines the principles of environmental stewardship and viable engineering design, which have long-term advantages to consumers and the sustainability agenda across the world. The

SolarVision Mower combines energy efficiency, affordability, and environmental protection; it has excellent potential for success in the business and the environment.

Timeline and Milestones

As indicated in the flowchart, the project begins in September- October 2025 with the manufacture of Solar Sled, Solar Panel, Battery and BMS and the commencement of the System Bench-Test of the LiDAR, GPS/RTK, YOLO and triangulation sensors. It transfers to the start of the Navigation System in November, 2025, composed of Arduino interfaces and early brevity of YOLO, LiDAR, and SLAM. This goes up to December 2025 that the navigation branches will converge and feasibility study would be conducted and the onboard navigation computer fed with validated information. The project begins the Physical design and iterations in January 2026 with the integration of the earlier created power and navigation subsystems into the emerging mechanical structure. The design cycling goes on to February 2026, to March 2026, when all the subsystems converge to the node of the Completed Project. Finally, the month of April-May 2026 will act as a buffer period where more debugging, testing, and re-work will be carried out after which the fully integrated system will then be finalized.

Time Period	Flowchart-Derived Milestone
Sept–Oct 2025	Begin Solar Sled, Solar Panel, Battery, BMS, and System Bench-Test
Nov 2025	Begin Navigation System (LiDAR → Arduino, GPS/RTK → Arduino, YOLO path, Triangulation path)
Dec 2025	Continue Navigation System; complete Test for Feasibility; feed results to On-board Computer
Jan 2026	Begin Physical Design + Iterations
Feb 2026	Continue Physical Design + Reiterations
March 2026	Integrate subsystems toward Completed Project
April–May 2026	Buffer Period (additional integration, testing, and adjustments)

Conclusion

SolarVision Mower is an innovative design of sustainable automation of residential landscapes. The design also incorporates solar energy, YOLO-based perception, and LiDAR navigation, making it focus on the sustainability, safety, and accessibility concerns mentioned in the previous sections directly. The project achieves these objectives due to novelty in the use of renewable energy sources, the use of AI-based precision, and the design to meet the needs of elderly and disabled people.

It is anticipated to be a complete prototype that is able to reliably mow autonomously with minimum human participation. The device has the potential of becoming a commercial in the emerging market of environmentally friendly smart devices that can lower the energy bill and further develop environmental objectives. SolarVision Mower is part of the larger decarbonization efforts; it promotes the UN Sustainable Development Goals (SDGs) concerning affordable clean energy and sustainable urban areas.

The research work of ECE 416 in the future will aim to run field testing, to develop the YOLO-LiDAR fusion algorithm, and to enhance the mobile connection. The project schedule makes the technical objectives as feasible as possible, and the continuous validation is in line with the IEEE and IEC standards of safety. In the end analysis, the SolarVision Mower serves the interests of homeowners, the environment, and the community, as it combines ethical responsibility with technological excellence, thus making it a feasible and socially responsible technology to live in the modern world.

Project Appendix

Necessary Components:

- Batteries and BMS:
  - 40 Samsung 18650 batteries, for 36V 10Ah battery pack in 10S 4P configuration
  - 3 12V 15Ah deep cycle lead acid batteries for stationary power storage at shed
  - 15A Lithium Ion BMS
- Solar and Power (stationary):
  - 3 12V 50-100W Solar panels, will decide depending on size constraints
  - MPPT Solar Charge Controller
- Power (on-board):
  - High-power relay array for safety, charging, and motor switching.
  - Custom PCB for drive, blade, and height motor controllers (all DC motors)
  - 36V to 12V buck converter for low power electronics
- Logic:
  - ESP32 S3 Dev-board
  - STM32N6x7xx soldered to custom PCB
  - 16MB PSRAM chip with Flexible Memory Controller interface (necessary to store YOLO weights)
  - 433MHz RF Module for communicating with charging dock/server.

Technical Documents:

- Fusion360 Documentation https://help.autodesk.com/view/fusion360/ENU/
  - We will be using Fusion360 to design all the physical components for this project. Required parts will then be either 3D printed out of TPU, or milled out of aluminum.
- KiCAD Documentation https://docs.kicad.org/
  - All PCB/Circuit Design (and some simulation) will be done in KiCAD. We will all need to learn and use KiCAD, and the documentation will thus be very important.
- STM32N6x7xx Datasheet and Reference manual. https://www.st.com/resource/en/datasheet/stm32n657a0.pdf

https://www.st.com/resource/en/reference_manual/rm0486-stm32n647657xx-armbased-32bit-mcus-stmicroelectronics.pdf

    - The datasheet and reference manual will be essential to the project's PCB design; specifically for pin configurations and finding footprint dimensions.
- STM32 ARM Cortex-M55 Programming Manual. https://www.st.com/resource/en/programming_manual/pm0273-stm32-cortexm55-mcus-programming-manual-stmicroelectronics.pdf
    - The programming manual for STM32's ARM Cortex-M55 line of processors will be essential to programming the MCU once completed.
- ESP32 Datasheet https://documentation.espressif.com/esp32-s3_datasheet_en.pdf
    - We will be using an ESP32 S3 for collecting data from sensors and the STM32's YOLO model, controlling the motors, and running the core pathfinding algorithm.
- Miscellaneous other datasheets for the Solar Panels, BMS, MPPT charge controller, and power electronics

Software Tools:

- Git: We will be using Git as the version control system for our project. The main repository will house all the 3D models, KiCAD files, code, and documentation for the project.
- KiCAD: To design the mower's custom PCB, and simulate various components, we will be using KiCAD. It will house the main ESP32 dev board via header pins and sockets, as well as the STM32 chip directly on the PCB.
- Fusion360: We will use Fusion360 to design all the physical components of the mower. Namely:
    - The mower's *body*, which will be welded out of 2x2 carbon steel tubing, and attached to the chassis of an existing mower
    - The mower's *attachments* for motors, sensors, and other peripherals. Prototypes for these attachments will be 3D printed out of PLA, and the final versions will be out of TPU with significant infill, for strength.
    - The mower's *electronics housing box*, which will house the main controller's PCB, the power electronics (relays, BMS, etc), as well as external LEDs as status indicators

Simulations:

- KiCAD: For hardware level simulations, such as passive/active circuits, amplifiers, voltage regulators, etc.
- QEMU: For low-level hardware emulation:
    - Espressif's fork of QEMU for the ESP32 to emulate code
    - QEMU's ARM Cortex-M55 emulator for STM32N6 emulations (not counting the neural accelerator)

Project Management Software:

- Git: We will use Git as the primary version control and project management software.

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